Choosing an element of each subset

Animath — Stage olympique de Sophia-Antipolis · 2006 · France

Let $A = \{1, 2, 3, 4, 5\}$. Count the functions $f : \mathfrak{P}(A) \to A$ such that :
— $\forall B \subseteq A,\ f(B) \in B$
— $\forall B, C \subseteq A,\ f(B \cup C) = f(B)$ or $f(C)$
Here $\mathfrak{P}(A)$ denotes the set of subsets of A.